

Day-8

IP Internet Protocol Location of device
MAC Media access Control Identification of device

MAC

- Media access control
- It is a physical address
- It is in hexa decimal format
- It contains 12 hexa decimal
- Each hexa contains 4 bit
- $12 \times 4 = 48$ bit length

NIC MAC

TC-CI-DE-33-8D-14

5-Subnet

Binary value
48 bit

24 bit 24 bit
Vendor id User id

IP

- Internet protocol
- It is a logical address
- It is in dotted decimal format
- It contains 4 octet
- Each octet contains 8 bit
- $8 \times 4 = 32$ bit length
- IP = network + host

IPv4 Structure

32 bit

0 0

0 0

0 0

0 0

0.0.0.0 Start byte

255.255.255.255 End byte

Classes
A B C D E

CLASS	RANGE	PORTION
A	0 - 127	N.H.H.H
B	128 - 191	N.N.H.H
C	192 - 223	N.N.N.H
D	224 - 239	Multicast
E	240 - 255	Reserved

Types of Communication

- unicast one $\rightarrow$ one
- multicast one $\rightarrow$ group
- Broadcast one $\rightarrow$ All

Invalid

Valid

10.0.0.10 - A - N.H.H.H

172.16.254.192 - B - N.N.H.H

192.0.0.192 - C - N.N.N.H

50.0.0.10
N.H.H.H

172.16.9.10
N.N.H.H

50.0.0.18

172.19.9.10

50.10.10.10

192.192.192.192
N.N.N.H

How many n/w bits?
24

How many host bits?
8

How many host octet?
1

How many n/w octet?
3

2 bit

2	1
0	0
0	1
1	0
1	1

3 bit

4	2	1
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

5 bit

4 bit

Thank-u